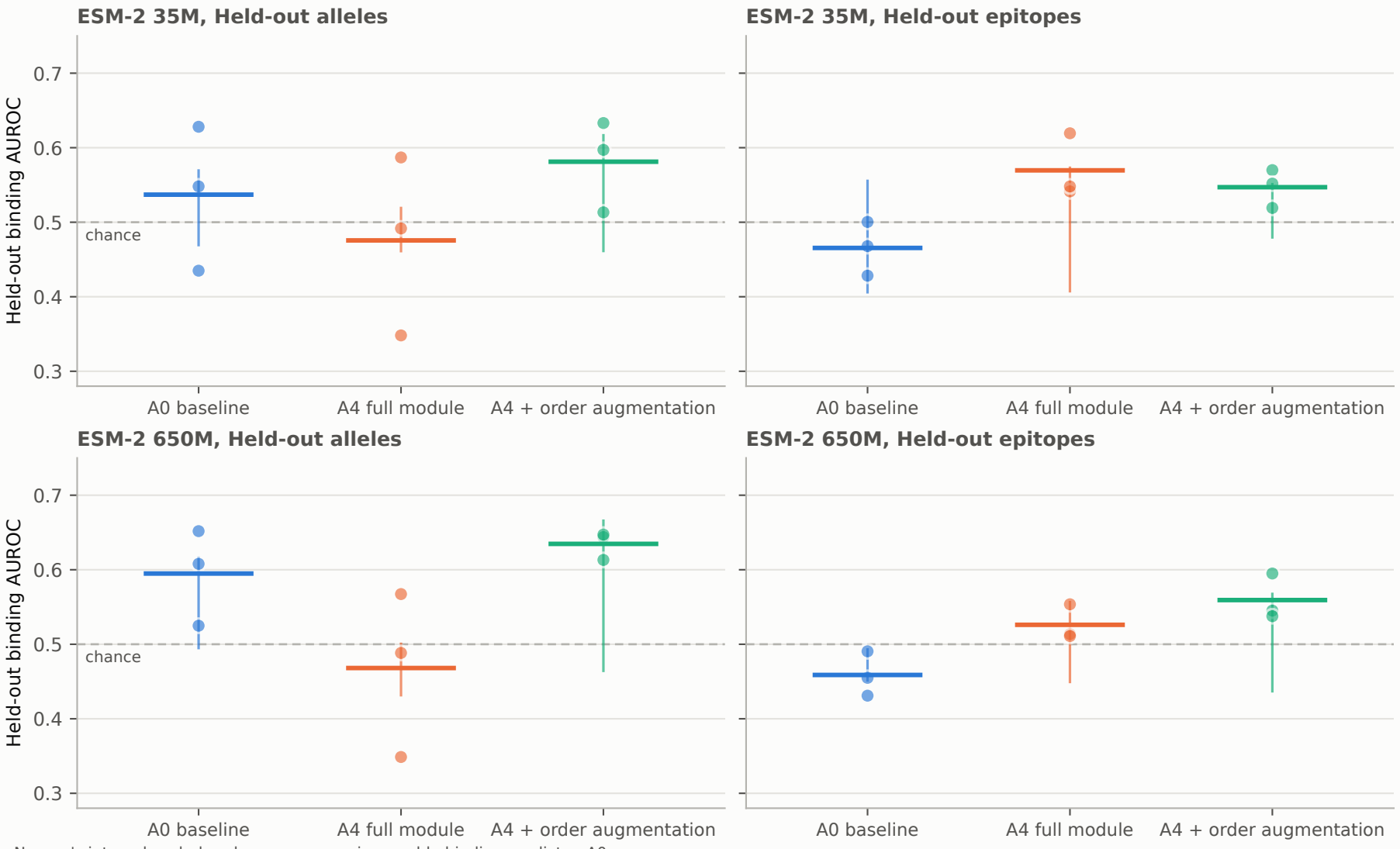
The chain-aware module does not improve held-out binding over a matched baseline

ESM-2 35M, ESM-2 650M. LoRA r=16, trainable parameters matched across arms, 300 steps, three seeds. Each dot is one seed; the bar is the seed mean, the thin line a cluster bootstrap over MHC alleles.



No arm's interval excludes chance, so none is a usable binding predictor. A0 on held-out epitopes sits below chance at both sizes. Between-seed spread swamps the between-arm differences here; fig8 pairs the seeds, which is what the design supports. Negatives are mismatched decoys, so AUROC is a weak proxy for binding, not a bound.